

K-FIELD DUAL-AXIS KINETIC MASS STARTING ORIENTATION

Canonical ICRS datum, directed axis polarities, zero-phase mass labeling, active rotation order, signed shaft rates, orientation sequence, Cell3 embedding, and code-stack definition

Zero-phase dual-axis apparatus at Earth center: $R = 0.5 \text{ m}$

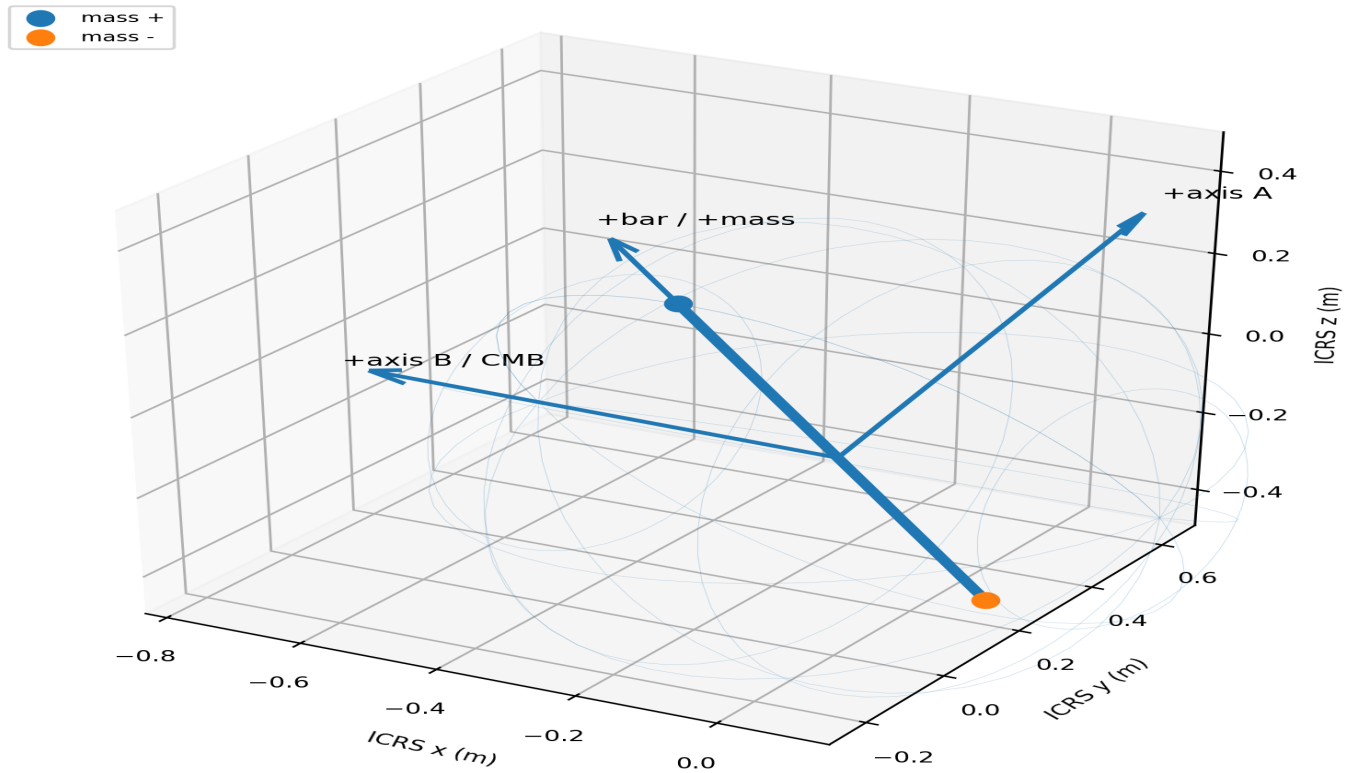


Figure 1. Dimensionally correct zero-phase apparatus at Earth center. The 1 m massless bar carries two endpoint masses at radius 0.5 m. The directed device axes are embedded in ICRS and axis B is locked to the Earth-CMB direction.

P. Dwayne Esterline, Principal Investigator

BS, Huntington University (1999)

MBA, Indiana Wesleyan University (2008)

Krytronx

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Frontispiece: authoritative Cell3 base cell, planes, and all nodes

Authoritative centered Cell3: all six faces and 27 half-step nodes

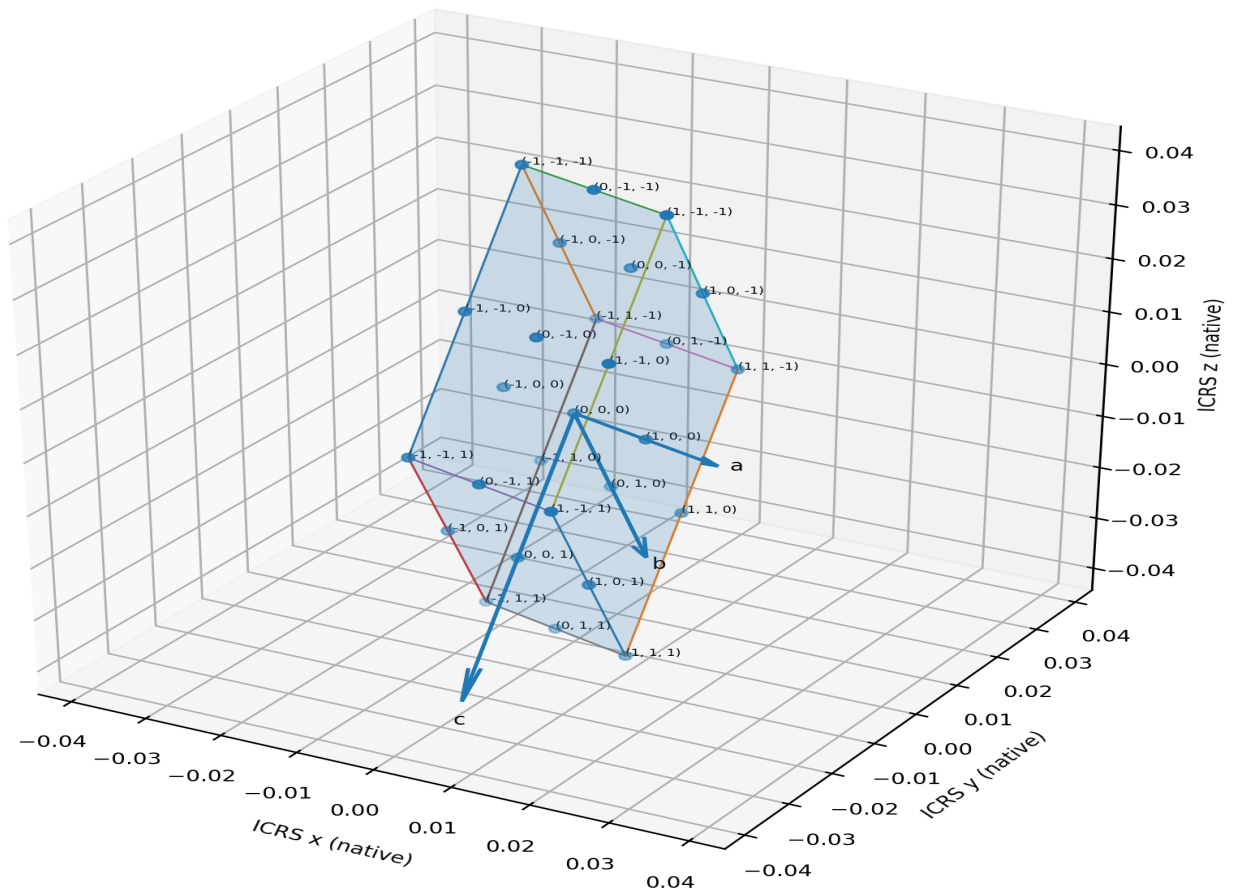


Figure 2. The centered primitive Cell3 is rendered directly from the authoritative ICRS basis. All six boundary faces, twelve edges, 27 half-step nodes, integer labels, and basis vectors are shown in native dimensions.

Cell3 central plane families and zero-phase device orientation

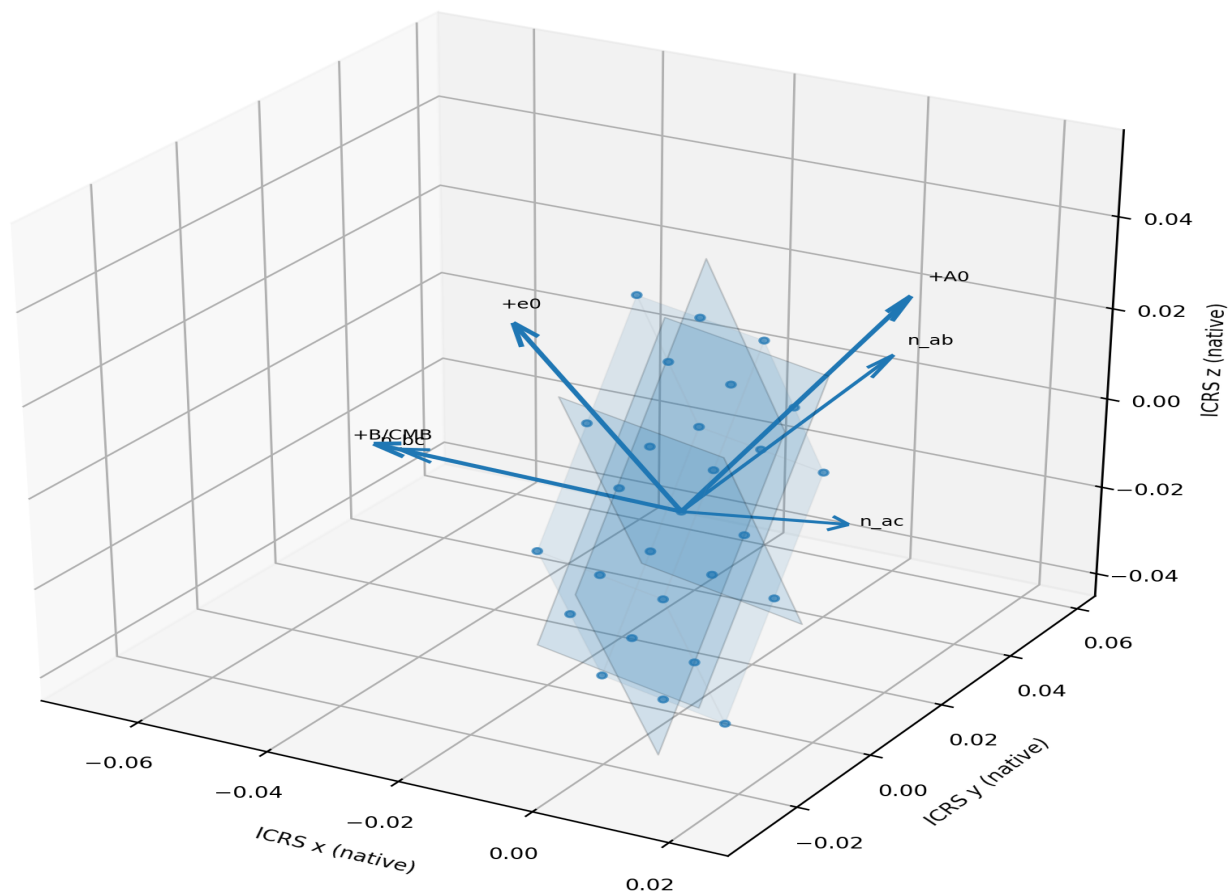


Figure 3. The three central face-plane families ab, ac, and bc, their fixed ICRS normals, and the complete zero-phase apparatus triad. The cell and planes retain native-length scaling; apparatus arrows are normalized orientation vectors.

Abstract

This reference removes every sign, axis, phase, and coordinate ambiguity from the initial configuration of the K-Field dual-axis kinetic-mass apparatus. The device origin is Earth center. A seasonal position datum places Earth center at one astronomical unit on the positive ICRS X axis, corresponding to RA=0 degrees and Dec=0 degrees. The device orientation is independent of that position datum.

At zero mechanical phase, the positive bar direction is $\mathbf{e}_0$, the positive inner-axis direction is $\mathbf{A}_0 = \mathbf{c}_{\text{hat}} \times \mathbf{e}_0$, and the positive outer-axis direction is $\mathbf{c}_{\text{hat}}$, the Earth-CMB traversal vector. These three unit vectors form a right-handed orthonormal device frame embedded in ICRS. The positive mass is the endpoint at $+R \mathbf{e}_0$; the negative mass is the antipodal endpoint at $-R \mathbf{e}_0$.

The outer assembly uses a positive right-hand rotation about $+\mathbf{c}_{\text{hat}}$. Viewed from the positive end of axis B toward the center, the outer motion is counterclockwise and immediately carries $\mathbf{e}_0$ toward $+\mathbf{A}_0$. The inner motion required by the exact path uses a negative right-hand rotation about the transported $+\mathbf{A}(\phi)$ axis. Viewed from the positive end of axis A toward the center, the inner motion is clockwise and immediately carries the positive endpoint from $+\mathbf{e}_0$ toward $+\mathbf{c}_{\text{hat}}$. The signed rate lock is $\omega_A = -2 \omega_B$.

The exact active compound rotation is $\mathbf{u}(\phi) = \mathbf{R}(\mathbf{c}_{\text{hat}}, \phi) \mathbf{R}(\mathbf{A}_0, -2\phi) \mathbf{e}_0$. Its equivalent direct form is $\mathbf{u}(\phi) = \cos(2\phi) \mathbf{e}(\phi) + \sin(2\phi) \mathbf{c}_{\text{hat}}$. The resulting sequence passes through $+\mathbf{e}_0$, $+\mathbf{c}_{\text{hat}}$, $-\mathbf{A}_0$, $-\mathbf{c}_{\text{hat}}$, $-\mathbf{e}_0$, $+\mathbf{c}_{\text{hat}}$, $+\mathbf{A}_0$, $-\mathbf{c}_{\text{hat}}$, and returns to $+\mathbf{e}_0$ over one outer cycle. The zero-phase derivative gate is $d\mathbf{u}/dt = \omega_B (\mathbf{A}_0 + 2\mathbf{c}_{\text{hat}})$. These definitions supply a complete code-stack orientation contract and a mechanically testable motor-direction specification.

Central result

The complete initial state is fixed by the tuple $(t_0, r_{C0}, \mathbf{e}_0, \mathbf{A}_0, \mathbf{c}_{\text{hat}}, r_{\text{plus0}}, r_{\text{minus0}}, \omega_B > 0, \omega_A = -2 \omega_B)$. This tuple determines the coordinate frame, axis polarity, mass labels, phase origin, matrix order, and both rotation directions without residual convention freedom.

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1. Coordinate datum and ICRS frame

All vectors are represented in the fixed International Celestial Reference System Cartesian frame. The positive X axis points to RA=0 degrees, Dec=0 degrees; the positive Y axis points to RA=90 degrees, Dec=0 degrees; and the positive Z axis points to the north celestial pole, Dec=+90 degrees.

```
X_ICRS = (1, 0, 0)
Y_ICRS = (0, 1, 0)
Z_ICRS = (0, 0, 1)
```

At the selected seasonal datum, the device origin is Earth center at

```
r_C(t0) = (149597870700, 0, 0) m
```

This is one astronomical unit on the positive ICRS X axis. It fixes an absolute position and seasonal phase reference. It does not rotate the device and does not redefine the Earth-CMB traversal direction.

The origin-centered device geometry is calculated from relative positions. Therefore, for orientation and rotor kinematics, the large astronomical translation and the meter-scale device geometry remain separate coordinate layers.

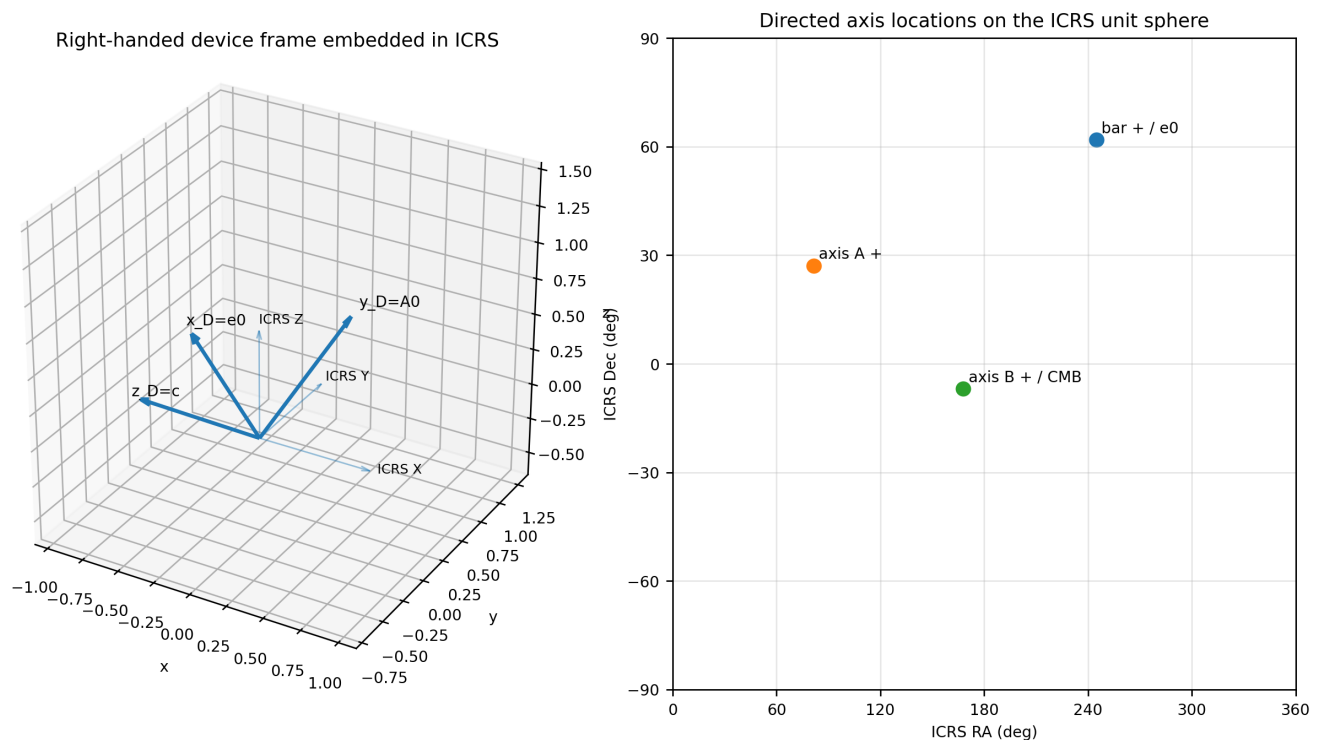


Figure 4. Left: the device triad embedded in ICRS. Right: the directed bar and axes expressed as ICRS RA and Dec locations on the unit sphere.

2. Directed zero-phase device triad

At $t=t_0$ and outer phase $\phi=0$, define the positive device axes

```
x_D = e0
y_D = A0
z_D = c_hat
```

with the authoritative ICRS values

```
e0 = (-0.200864884110, -0.426231479950, 0.882031758970)
A0 = (0.131366829113, 0.880530254232, 0.455422032396)
c = (-0.970770855836, 0.207348009111, -0.120874929482)
```

Axis A is not selected independently. It is defined by the right-handed cross product

```
A0 = c_hat cross e0
```

and therefore

$$\mathbf{e}_0 \text{ cross } \mathbf{A}_0 = \mathbf{c}_{\text{hat}}.$$

This makes $(\mathbf{e}_0, \mathbf{A}_0, \mathbf{c}_{\text{hat}})$ a right-handed orthonormal frame. The orientation matrix that maps device coordinates into ICRS is

$$\begin{aligned} \mathbf{R}_{\text{ICRS_from_D}}(\theta) &= [\mathbf{e}_0 \quad \mathbf{A}_0 \quad \mathbf{c}_{\text{hat}}] \\ &= [\begin{array}{ccc} -0.200864884110 & 0.131366829113 & -0.970770855836 \\ -0.426231479950 & 0.880530254232 & 0.207348009111 \\ 0.882031758970 & 0.455422032396 & -0.120874929482 \end{array}] \end{aligned}$$

For any device-coordinate vector $\mathbf{q}_D$, its ICRS representation is $\mathbf{q}_{\text{ICRS}} = \mathbf{R}_{\text{ICRS_from_D}} \mathbf{q}_D$. The inverse is the transpose because the matrix is orthogonal.

$$\begin{aligned} \mathbf{R}^T \mathbf{R} &= \mathbf{I} \\ \det(\mathbf{R}) &= +1 \\ \mathbf{q}_D &= \mathbf{R}^T \mathbf{q}_{\text{ICRS}} \end{aligned}$$

Object	Directed ICRS vector	RA (deg)	Dec (deg)	Role
+bar / +e0	(-0.200864884110, -0.426231479950, 0.882031758970)	244.767479396	61.888432983	positive endpoint direction at phase zero
+axis A / +A0	(0.131366829113, 0.880530254232, 0.455422032396)	81.514594988	27.092091822	inner shaft polarity and transported rotation axis
+axis B / +c	(-0.970770855836, 0.207348009111, -0.120874929482)	167.943299880	-6.942599922	outer shaft polarity and Earth-CMB direction

3. Axis lines and endpoint labels

Each mechanical axis is a directed line through the Earth-center device origin. The positive direction is part of the definition and must not be discarded.

$$\begin{aligned} \mathbf{L}_B(\theta) &= \{ \mathbf{r}_C(t_0) + s \mathbf{c}_{\text{hat}} : s \text{ in } \mathbf{R} \} \\ \mathbf{L}_A(\theta) &= \{ \mathbf{r}_C(t_0) + s \mathbf{A}_0 : s \text{ in } \mathbf{R} \} \\ \mathbf{L}_{\text{bar}}(\theta) &= \{ \mathbf{r}_C(t_0) + s \mathbf{e}_0 : s \text{ in } \mathbf{R} \} \end{aligned}$$

The positive endpoint is permanently labeled mass +. At zero phase it lies in the +e0 direction. The antipodal endpoint is mass -.

$$\begin{aligned} \mathbf{r}_{\text{plus}}(t_0) &= \mathbf{r}_C(t_0) + \mathbf{R} \mathbf{e}_0 \\ \mathbf{r}_{\text{minus}}(t_0) &= \mathbf{r}_C(t_0) - \mathbf{R} \mathbf{e}_0 \end{aligned}$$

For $R=0.5$ m, the relative endpoint positions are

$$\begin{aligned} \mathbf{r}_{\text{plus}} - \mathbf{r}_C &= (-0.100432442055, -0.213115739975, 0.441015879485) \text{ m} \\ \mathbf{r}_{\text{minus}} - \mathbf{r}_C &= (0.100432442055, 0.213115739975, -0.441015879485) \text{ m} \end{aligned}$$

The labels follow the physical masses continuously. They are not reassigned when a mass crosses an axis, reaches a pole, or becomes the geometrically opposite endpoint.

4. Active rotation convention

All rotations are active rotations of vectors in the fixed ICRS frame. The rotation matrix about a directed unit axis $\mathbf{k}$ through signed angle θ is the Rodrigues matrix

$$\begin{aligned} \mathbf{R}(\mathbf{k}, \theta) &= \mathbf{I} \cos(\theta) \\ &\quad + (1 - \cos(\theta)) \mathbf{k} \mathbf{k}^T \\ &\quad + \sin(\theta) [\mathbf{k} \text{ cross}] \end{aligned}$$

where $[\mathbf{k} \text{ cross}] \mathbf{v} = \mathbf{k} \text{ cross } \mathbf{v}$. Positive θ follows the right-hand rule about the positive axis. Viewed from the positive axis end toward the device center, positive rotation is counterclockwise.

This convention fixes the signs used by every matrix, encoder, angular-velocity vector, and motor command. Replacing an axis by its negative requires replacing the corresponding signed angle by its negative. Reversing both leaves the physical rotation unchanged, but mixing conventions does not.

5. Outer-axis rotation direction

The outer phase is

$$\phi = \omega_B (t - t_0), \quad \omega_B > 0$$

and the outer angular-velocity vector is

$$\omega_{B_vector} = +\omega_B \hat{c}.$$

The outer assembly therefore rotates by a positive right-hand angle about +axis B. Viewed from the + $\hat{c}$ end toward Earth center, the motion is counterclockwise.

$$R_B(\phi) = R(\hat{c}, +\phi)$$

The transported transverse basis is

$$\begin{aligned} e(\phi) &= R_B(\phi)e_0 \\ &= \cos(\phi)e_0 + \sin(\phi)A_0 \\ A(\phi) &= R_B(\phi)A_0 \\ &= -\sin(\phi)e_0 + \cos(\phi)A_0 \end{aligned}$$

Immediately after t_0 , positive outer rotation carries e_0 toward $+A_0$. This single mapping is the simplest practical sign check for the outer motor and encoder.

Outer rotation: positive right-hand rotation about +axis B

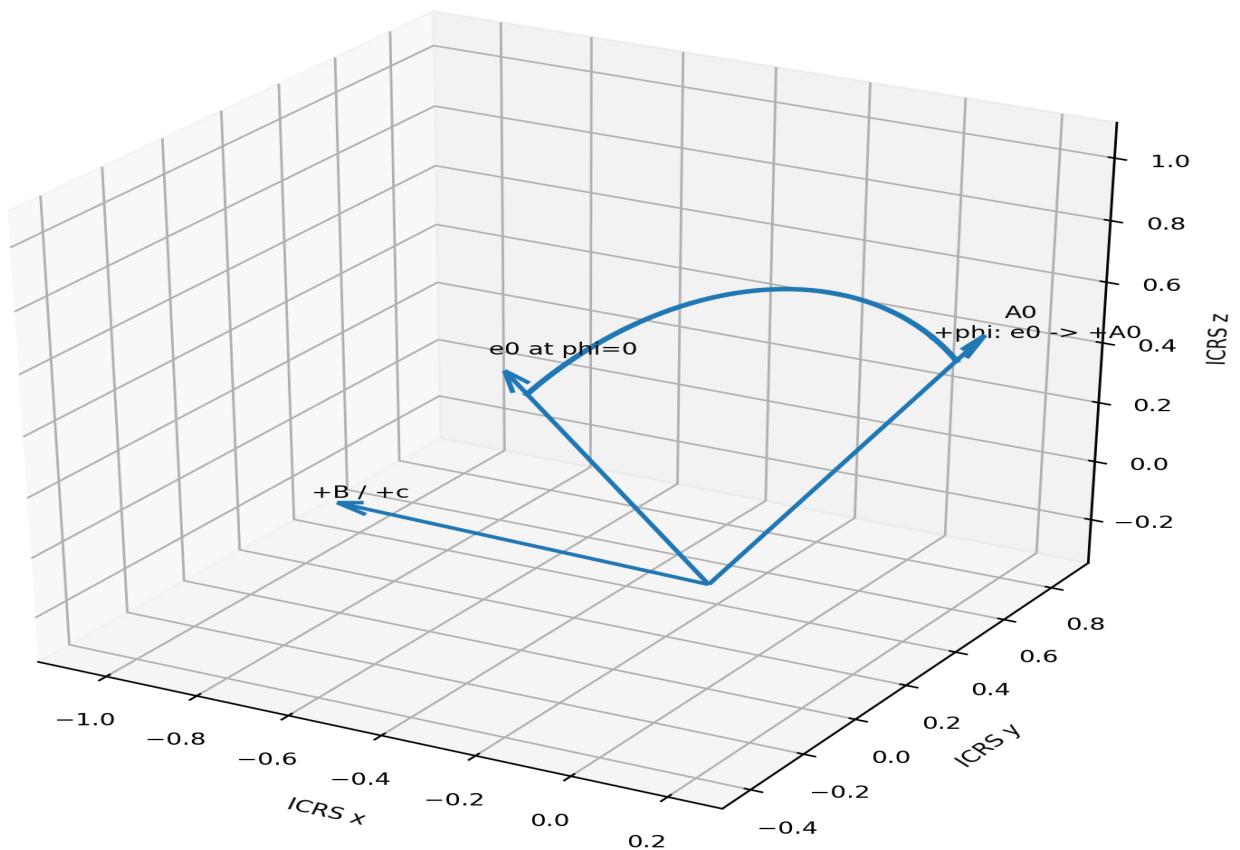


Figure 5. Positive outer rotation about the directed +CMB axis. The quarter-turn arc shows e_0 advancing toward $+A_0$.

6. Inner-axis rotation direction

The exact endpoint path requires the bar to rotate through signed inner phase

$$\beta = -2\phi$$

about the positive transported axis $A(\phi)$. Therefore the inner angular-velocity vector is

$$\omega_{A_vector}(\phi) = -2\omega_B A(\phi).$$

Viewed from the positive end of $A(\phi)$ toward the center, this is clockwise rotation. At zero phase it carries the positive bar direction $+e_0$ toward $+c_{\hat{c}}$.

The negative sign is essential. A positive inner angle about $+A_0$ would carry e_0 toward $-c_{\text{hat}}$ under the declared right-handed frame and would generate the mirror path with reversed pole order.

Inner rotation: negative right-hand rotation about $+axis\ A$

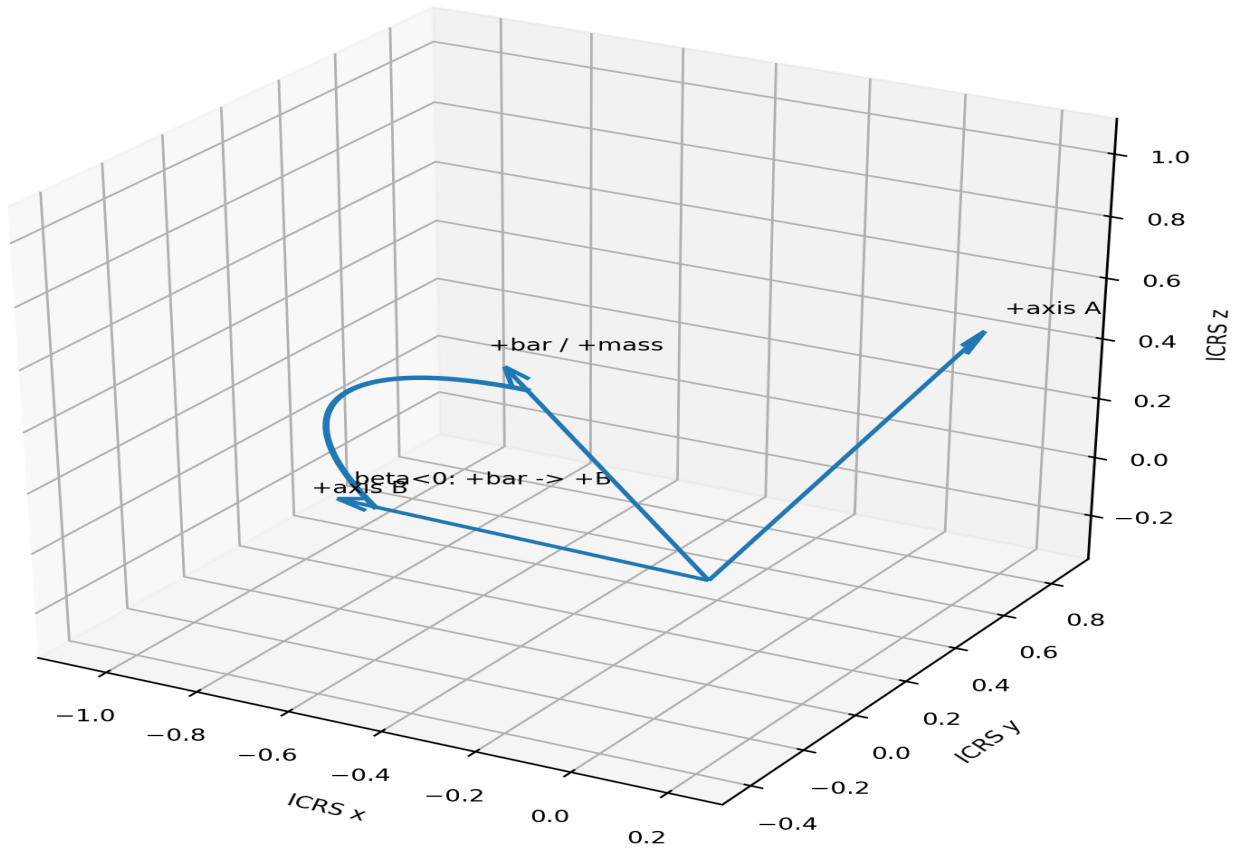


Figure 6. Required inner rotation at zero phase. The signed angle is negative about $+A_0$, so the positive endpoint advances from $+e_0$ toward $+c_{\text{hat}}$.

The recommended code convention retains the geometrically defined $+A(\phi)$ axis and stores a negative signed rate. An alternate motor convention may define the positive motor shaft as $-A(\phi)$ and store a positive rate, but the two conventions must never be mixed.

```
recommended: axis_A_positive = +A(phi), omega_A_signed = -2 omega_B
alternate:   motor_axis_positive = -A(phi), omega_motor = +2 omega_B
```

7. Compound rotation order

The exact active compound rotation is

$$u(\phi) = R(c_{\text{hat}}, \phi) R(A_0, -2\phi) e_0.$$

Matrix multiplication acts from right to left. First rotate the initial bar through -2ϕ about the initial $+A_0$ axis. Then carry the complete inner assembly through ϕ about $+c_{\text{hat}}$. Because the rotations are about different axes, they do not commute and their order cannot be reversed.

The same geometry can be evaluated with the transported basis

$$u(\phi) = \cos(2\phi)e(\phi) + \sin(2\phi)c_{\text{hat}}.$$

These two expressions must agree numerically at every phase and provide an independent implementation cross-check.

Active matrix order at $\phi = 25^\circ$: $u = R_c(+\phi) R_{A0}(-2\phi) e_0$
 Step 1: $R_{A0}(-2\phi)$ Step 2: $R_c(+\phi)$ applied to complete inner state

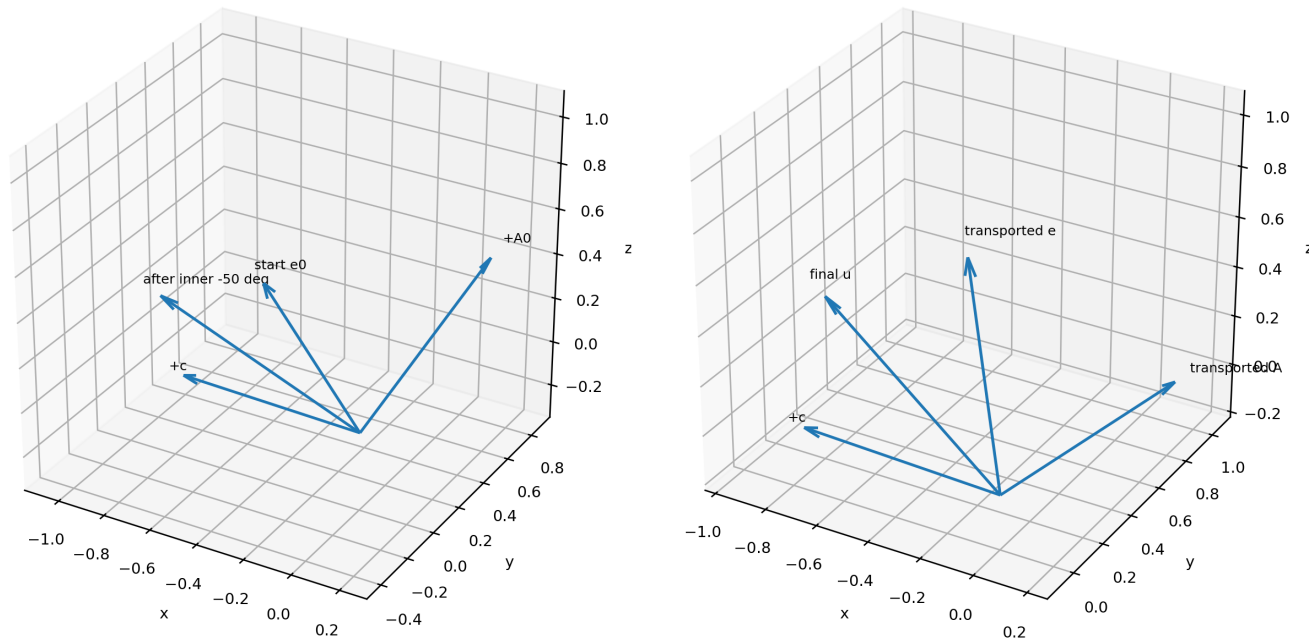


Figure 7. Active rotation order at a sample outer phase of 25 degrees. The inner -50 degree rotation is applied first; the outer +25 degree rotation then transports the complete state.

8. Zero-phase derivative gate

At $\phi=0$, the positive endpoint direction is

$$u(\theta)=e\theta.$$

Differentiating the direct path with respect to outer phase gives

$$\begin{aligned} u_{\phi} = & -2 \sin(2\phi)e(\phi) \\ & + \cos(2\phi)f(\phi) \\ & + 2 \cos(2\phi)c_{\text{hat}} \end{aligned}$$

where

$$f(\phi)=-\sin(\phi)e\theta+\cos(\phi)A\theta.$$

At zero phase

$$u_{\phi}(\theta)=A\theta+2c_{\text{hat}}$$

and therefore

$$du/dt \text{ at } t\theta = \omega_B(A\theta+2c_{\text{hat}}).$$

The same result must be obtained from the two signed angular velocities:

$$du/dt = [\omega_B c_{\text{hat}} - 2 \omega_B A(\phi)] \text{ cross } u(\phi).$$

At $t\theta$, $c_{\text{hat}} \text{ cross } e\theta=A\theta$ and $A\theta \text{ cross } e\theta=-c_{\text{hat}}$, so the outer contribution is $+\omega_B A\theta$ and the inner contribution is $+2\omega_B c_{\text{hat}}$.

Zero-phase derivative gate: $du/dt = \omega_B(A_0 + 2c)$

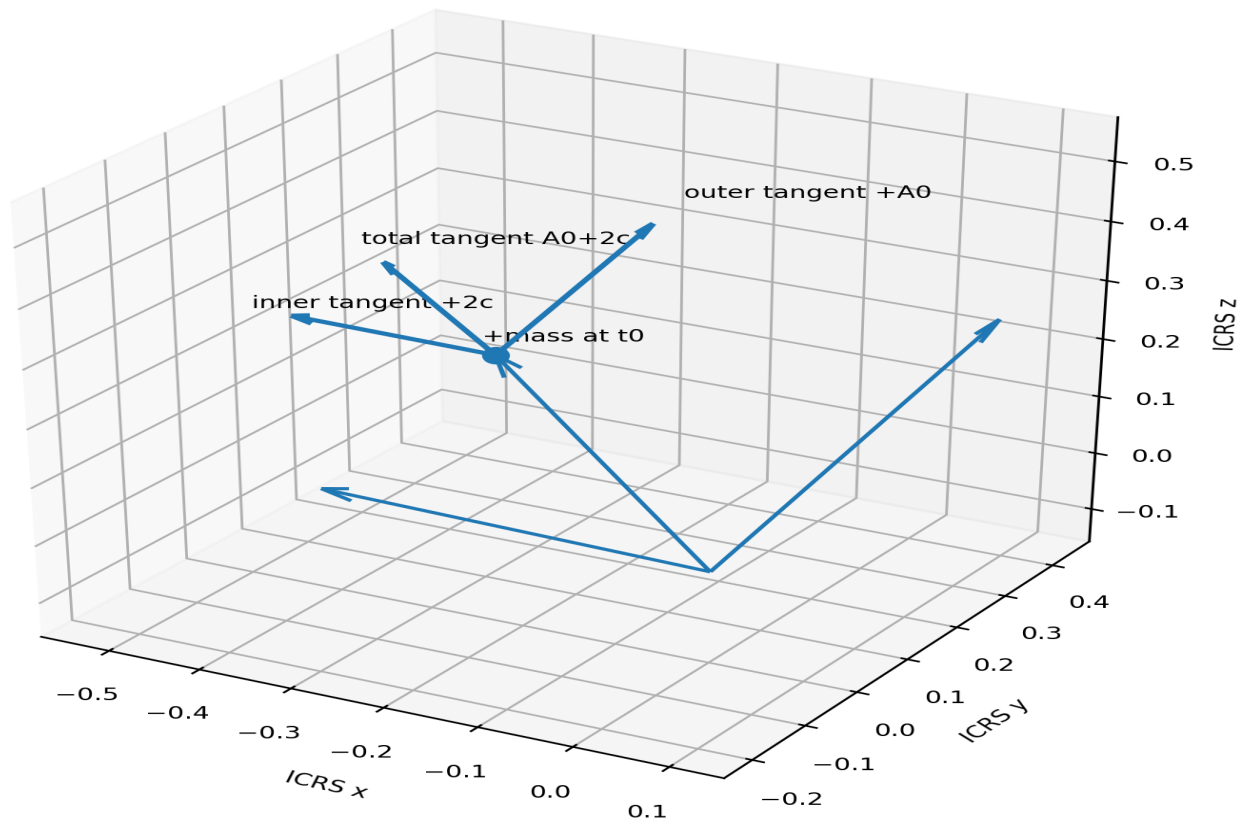


Figure 8. Zero-phase tangent decomposition. The outer tangent points along $+A_0$, the inner tangent points along $+2c_{\text{hat}}$, and their vector sum is the required path tangent.

Primary acceptance test. A physical or numerical implementation is correctly signed only when the measured or computed initial endpoint velocity points along A_0+2c_{hat} for $\omega_B > 0$.

9. Complete orientation sequence

Over one positive outer cycle, the positive endpoint follows the exact sequence below. The inner signed phase continues monotonically in the negative direction.

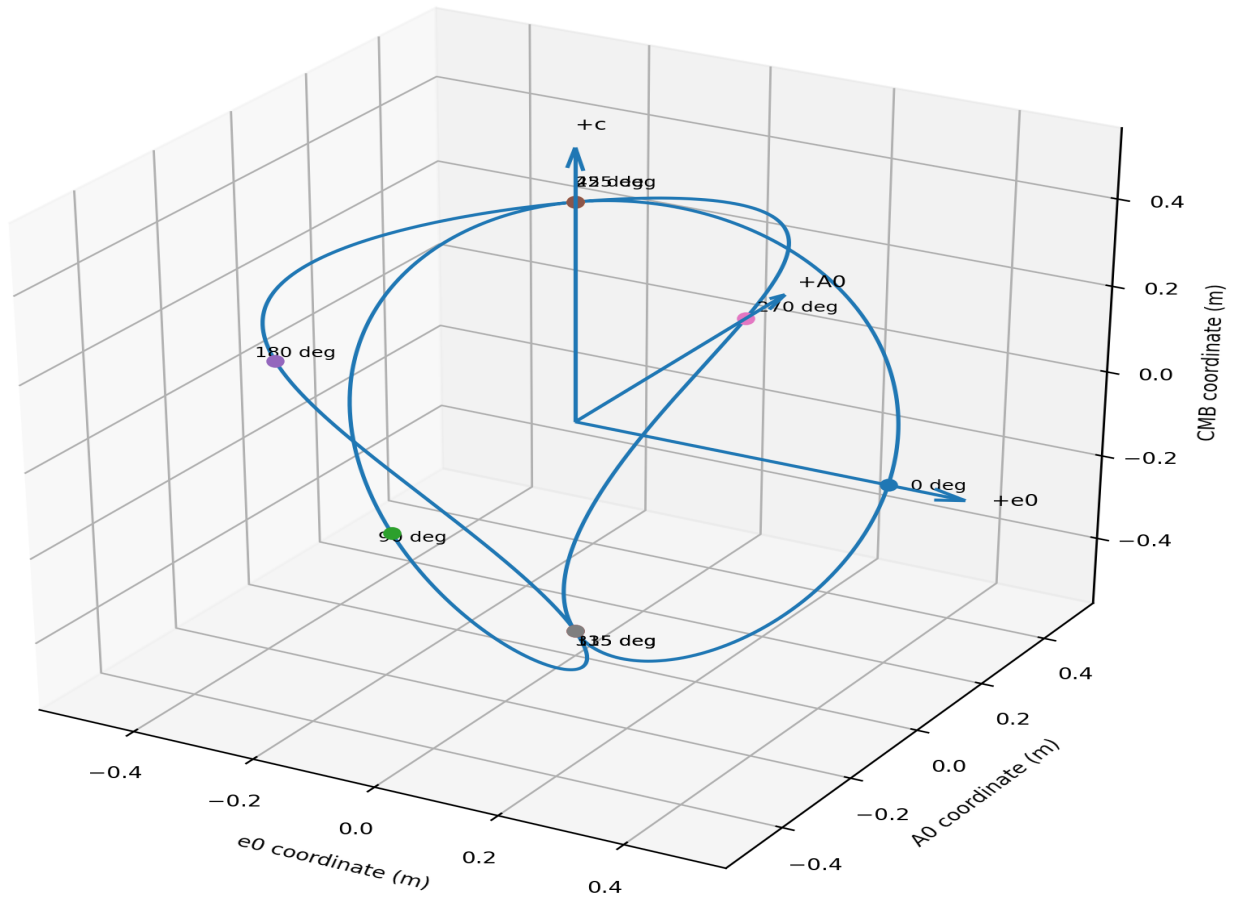
Exact positive-endpoint sequence on the $R = 0.5$ m sphere

Figure 9. The exact positive-endpoint path in the local device frame. The repeated visits to $+c_{\text{hat}}$ and $-c_{\text{hat}}$ occur at different outer phases and belong to the same labeled mass.

Outer phase	Inner signed phase	Positive endpoint direction	Geometric state
0 deg	0 deg	$+e_0$	initial equatorial radius
45 deg	-90 deg	$+c_{\text{hat}}$	positive CMB-axis centerline
90 deg	-180 deg	$-A_0$	equatorial radius
135 deg	-270 deg	$-c_{\text{hat}}$	negative CMB-axis centerline
180 deg	-360 deg	$-e_0$	opposite equatorial radius
225 deg	-450 deg	$+c_{\text{hat}}$	positive CMB-axis centerline
270 deg	-540 deg	$+A_0$	equatorial radius
315 deg	-630 deg	$-c_{\text{hat}}$	negative CMB-axis centerline
360 deg	-720 deg	$+e_0$	cycle closure

An implementation that produces a different ordered sequence has changed at least one of four items: axis polarity, inner-rate sign, matrix order, or phase definition.

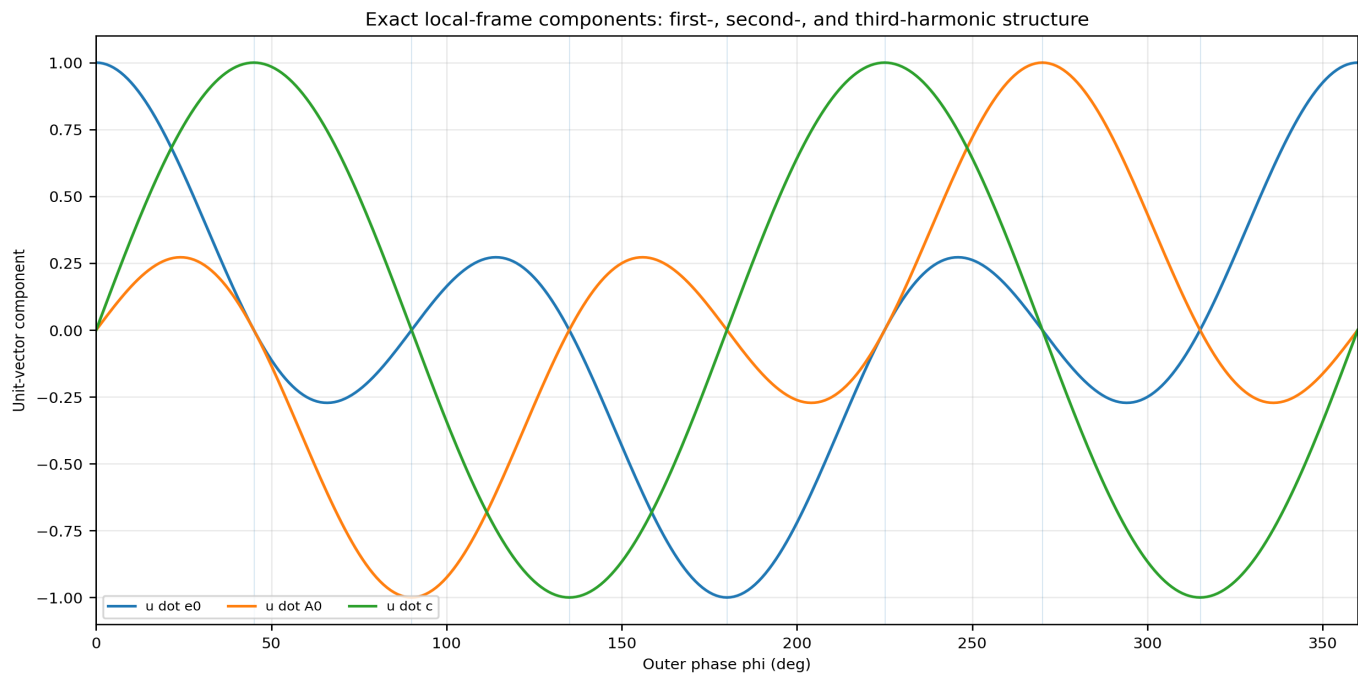


Figure 10. Local-frame components of the positive endpoint. The CMB component is $\sin(2\phi)$; the transverse components contain the first and third harmonics generated by the compound rotation.

10. Motor, encoder, and assembly implementation

The mathematical axis polarity must be transferred to physical shaft markings. Mark the positive end of the outer shaft in the $+c_{\hat{}}$ direction and the positive end of the inner shaft in the $+A_0$ direction at t_0 . Mark the bar endpoint carrying mass $+$ in the $+e_0$ direction.

Channel	Positive direction	Required command for $\omega_B > 0$	Visual check from positive end
Outer shaft B	$+c_{\hat{}}$	positive angle $+\phi$; angular velocity $+\omega_B c_{\hat{}}$	counterclockwise; e_0 advances toward $+A_0$
Inner shaft A	$+A(\phi)$	negative angle -2ϕ ; angular velocity $-2\omega_B A(\phi)$	clockwise; $+mass$ advances from $+e_0$ toward $+c_{\hat{}}$ at t_0
Outer encoder	increases with $+\phi$	one revolution per outer cycle	0,45,90,...,360 deg sequence
Inner encoder	signed $\beta = -2\phi$	decreases two revolutions per outer cycle	0,-90,-180,...,-720 deg
Mass label	mass $+$ fixed to $+e_0$ endpoint at t_0	never relabel during operation	returns to same endpoint at 360 deg

At the declared unit-rate normalization, $\omega_B = 1$ rad/s corresponds to 9.549296586 RPM. The inner signed rate is -2 rad/s, or -19.098593171 RPM. The negative RPM means clockwise when viewed from the positive A shaft end.

The inner shaft is transported by the outer rotation. Consequently, its ICRS direction is time dependent even though its body-fixed direction within the outer assembly is fixed.

$$A(\phi) = R(c_{\hat{}}, +\phi)A_0$$

11. Code-stack canonical definition

The following orientation contract is sufficient for direct implementation. All vectors are column vectors in ICRS unless explicitly labeled as device coordinates.

```
ICRS axes:
+X = RA 0 deg, Dec 0 deg
+Y = RA 90 deg, Dec 0 deg
+Z = Dec +90 deg

datum at t0:
r_C0 = (1 AU, 0, 0)

directed initial axes:
x_D = e0
z_D = c_hat
```

```

y_D = cross(z_D,x_D)

mass labels:
  r_plus_0 = r_C0 + R*x_D
  r_minus_0 = r_C0 - R*x_D

phase:
  phi = omega_B*(t-t0)

outer rotation:
  R_B = Rodrigues(z_D,+phi)
  e_phi = R_B@x_D
  A_phi = R_B@y_D
  omega_B_vec = +omega_B*z_D

inner rotation:
  beta = -2*phi
  omega_A_vec = -2*omega_B*A_phi

positive endpoint:
  u = R_B @ Rodrigues(y_D,beta) @ x_D

equivalent direct path:
  u = cos(2*phi)*e_phi + sin(2*phi)*z_D

endpoints:
  r_plus = r_C(t) + R*u
  r_minus = r_C(t) - R*u

```

The complete zero-phase state is

```

(t0, r_C0, e0, A0, c_hat, r_plus0, r_minus0,
omega_B>0, omega_A_signed=-2 omega_B).

```

No Euler-angle convention is required. The orientation matrix, directed axes, signed rates, and active Rodrigues rotations fully determine the state without gimbal singularities or order ambiguity.

12. Validation gates and failure diagnostics

A production code stack should reject the orientation input unless the following frame conditions are satisfied:

```

||e0||==||A0||==||c_hat||=1
e0 dot A0 = e0 dot c_hat = A0 dot c_hat = 0
A0 = c_hat cross e0
det[e0,A0,c_hat] = +1

```

At every phase, verify

```

e(phi) dot c_hat = 0
A(phi) dot c_hat = 0
e(phi) dot A(phi) = 0
||u(phi)|| = 1
R(c_hat,+phi)R(A0,-2phi)e0
= cos(2phi)e(phi)+sin(2phi)c_hat

```

At zero phase, verify

```

u(0)=e0
u_phi(0)=A0+2c_hat
du/dt=[omega_B c_hat-2omega_B A0] cross e0

```

At the cardinal phases, verify the complete sequence listed in Section 9. These gates diagnose the common implementation failures:

Failure	Observed symptom	Correction
Inner sign reversed	first pole is -c_hat instead of +c_hat	set beta=-2phi or reverse inner motor direction
Outer sign reversed	e0 initially advances toward -A0	set outer angle to +phi about +c_hat
Matrix order reversed	path fails direct closed-form comparison	apply R_A0 first, then R_c: R_c R_A0 e0
A axis polarity reversed only	handedness determinant becomes -1 or inner sign appears inconsistent	recompute A0=c_hat cross e0

Failure	Observed symptom	Correction
Mass labels swapped	initial +mass appears at -R e0	bind mass + to +R e0 at t0
Passive rotation used as active	both signs appear mirrored	use active Rodrigues matrices on vectors in fixed ICRS

13. Relationship to the Cell3 structure

The apparatus frame and Cell3 basis occupy the same fixed ICRS coordinates. The Cell3 basis vectors a, b, and c are not the device axes e0, A0, and c_hat. They are separate geometric objects represented in the same reference frame.

```
B_Cell3 = [a b c]
R_device = [e0 A0 c_hat]
```

The common frame permits direct projection of device axes into Cell3 coefficient coordinates without changing orientation conventions:

```
q_Cell3 = B_Cell3^-1 q_ICRS.
```

Axis B is fixed to the Earth-CMB direction, while the Cell3 planes retain their authoritative fixed normals. Therefore the zero-phase apparatus orientation specifies exactly how the bar and both rotation axes intersect the three Cell3 plane families at the initial datum.

The Cell3 base cell, its six face planes, and all 27 centered half-step nodes are reproduced in the frontispiece and appendices so the starting orientation can be reconstructed without external geometric assumptions.

Appendix A. Complete Cell3 boundary vertices and face planes

#	i	j	k	ICRS x	ICRS y	ICRS z
0	-0.5	-0.5	-0.5	-0.010974902500	0.007385983127	0.039661911012
1	-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443559
2	-0.5	0.5	-0.5	-0.008776156532	0.020288153386	0.005595409476
3	-0.5	0.5	0.5	-0.009406536017	-0.003969568631	-0.037328945095
4	0.5	-0.5	-0.5	0.009406536017	0.003969568631	0.037328945095
5	0.5	-0.5	0.5	0.008776156532	-0.020288153386	-0.005595409476
6	0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443559
7	0.5	0.5	0.5	0.010974902500	-0.007385983127	-0.039661911012

Family	Area native^2	Included angle	Unit normal ICRS	Boundary d
A_ab	7.547217896726e-04	83.933483012 deg	(0.1940922008, 0.9131771688, 0.3583792348)	0.018828573500
A_ac	1.011255757688e-03	80.448130018 deg	(0.0890525941, 0.8665767693, -0.4910339483)	0.014052166903
A_bc	1.385784431733e-03	50.363231814 deg	(-0.9959651903, 0.0836021625, -0.0326192906)	0.010254361619

Appendix B. Complete 27-node centered half-step registry

#	2i	2j	2k	lambda	ICRS x	ICRS y	ICRS z
0	-1	-1	-1	-3	-0.010974902500	0.007385983127	0.039661911012
1	-1	-1	0	-2	-0.011290092243	-0.004742877881	0.018199733726
2	-1	-1	1	-1	-0.011605281985	-0.016871738890	-0.003262443559
3	-1	0	-1	-2	-0.009875529516	0.013837068257	0.022628660244
4	-1	0	0	-1	-0.010190719258	0.001708207248	0.001166482958
5	-1	0	1	0	-0.010505909001	-0.010420653760	-0.020295694327
6	-1	1	-1	-1	-0.008776156532	0.020288153386	0.005595409476
7	-1	1	0	0	-0.009091346274	0.008159292378	-0.015866767810
8	-1	1	1	1	-0.009406536017	-0.003969568631	-0.037328945095
9	0	-1	-1	-2	-0.000784183241	0.005677775879	0.038495428054
10	0	-1	0	-1	-0.001099372984	-0.006451085129	0.017033250768
11	0	-1	1	0	-0.001414562727	-0.018579946138	-0.004428926518
12	0	0	-1	-1	0.000315189743	0.012128861009	0.021462177286
13	0	0	0	0	0.000000000000	0.000000000000	0.000000000000
14	0	0	1	1	-0.000315189743	-0.012128861009	-0.021462177286
15	0	1	-1	0	0.001414562727	0.018579946138	0.004428926518
16	0	1	0	1	0.001099372984	0.006451085129	-0.017033250768
17	0	1	1	2	0.000784183241	-0.005677775879	-0.038495428054
18	1	-1	-1	-1	0.009406536017	0.003969568631	0.037328945095
19	1	-1	0	0	0.009091346274	-0.008159292378	0.015866767810
20	1	-1	1	1	0.008776156532	-0.020288153386	-0.005595409476
21	1	0	-1	0	0.010505909001	0.010420653760	0.020295694327
22	1	0	0	1	0.010190719258	-0.001708207248	-0.001166482958
23	1	0	1	2	0.009875529516	-0.013837068257	-0.022628660244
24	1	1	-1	1	0.011605281985	0.016871738890	0.003262443559
25	1	1	0	2	0.011290092243	0.004742877881	-0.018199733726
26	1	1	1	3	0.010974902500	-0.007385983127	-0.039661911012

Appendix C. Complete one-degree orientation ledger

The table records the positive endpoint for outer phase 0 through 360 degrees. Local components are resolved on the fixed zero-phase frame (e0,A0,c_hat). The repeated 360-degree row is the closure state.

phi deg	beta deg	u.e0	u.A0	u.c	u ICRS x	u ICRS y	u ICRS z
0	0	1.00000000	0.00000000	0.00000000	-0.20086488	-0.42623148	0.88203176
1	-2	0.99923861	0.01744177	0.03489950	-0.23230009	-0.40331260	0.88508509
2	-4	0.99695636	0.03481448	0.06975647	-0.26339761	-0.37981511	0.88677065
3	-6	0.99315894	0.05204925	0.10452846	-0.29412640	-0.35581089	0.88706723
4	-8	0.98785583	0.06907761	0.13917310	-0.32445623	-0.33137306	0.88595714
5	-10	0.98106026	0.08583165	0.17364818	-0.35435771	-0.30657580	0.88342622
6	-12	0.97278921	0.10224427	0.20791169	-0.38380230	-0.28149414	0.87946395
7	-14	0.96306329	0.11824930	0.24192190	-0.41276229	-0.25620378	0.87406345
8	-16	0.95190676	0.13378177	0.27563736	-0.44121087	-0.23078087	0.86722152
9	-18	0.93934743	0.14877802	0.30901699	-0.46912211	-0.20530184	0.85893865
10	-20	0.92541658	0.16317591	0.34202014	-0.49647098	-0.17984316	0.84921906
11	-22	0.91014888	0.17691502	0.37460659	-0.52323335	-0.15448114	0.83807067
12	-24	0.89358230	0.18993678	0.40673664	-0.54938599	-0.12929179	0.82550510
13	-26	0.87575801	0.20218467	0.43837115	-0.57490661	-0.10435053	0.81153765
14	-28	0.85672028	0.21360436	0.46947156	-0.59977380	-0.07973206	0.79618728
15	-30	0.83651630	0.22414387	0.50000000	-0.62396711	-0.05551012	0.77947654
16	-32	0.81519615	0.23375373	0.52991926	-0.64746697	-0.03175732	0.76143154
17	-34	0.79281257	0.24238713	0.55919290	-0.67025475	-0.00854494	0.74208190
18	-36	0.76942088	0.25000000	0.58778525	-0.69231272	0.01405726	0.72146066
19	-38	0.74507881	0.25655121	0.61566148	-0.71362407	0.03598124	0.69960420
20	-40	0.71984631	0.26200263	0.64278761	-0.73417287	0.05716082	0.67655217
21	-42	0.69378546	0.26631929	0.66913061	-0.75394411	0.07753188	0.65234737
22	-44	0.66696025	0.26946943	0.69465837	-0.77292365	0.09703256	0.62703566
23	-46	0.63943640	0.27142465	0.71933980	-0.79109824	0.11560337	0.60066583
24	-48	0.61128123	0.27215994	0.74314483	-0.80845548	0.13318736	0.57328951
25	-50	0.58256342	0.27165378	0.76604444	-0.82498386	0.14973030	0.54496098
26	-52	0.55335287	0.26988823	0.78801075	-0.84067267	0.16518080	0.51573710
27	-54	0.52372049	0.26684892	0.80901699	-0.85551208	0.17949045	0.48567711
28	-56	0.49373803	0.26252517	0.82903757	-0.86949305	0.19261395	0.45484251
29	-58	0.46347783	0.25690996	0.84804810	-0.88260735	0.20450923	0.42329687
30	-60	0.43301270	0.25000000	0.86602540	-0.89484756	0.21513756	0.39110570
31	-62	0.40241567	0.24179573	0.88294759	-0.90620703	0.22446365	0.35833628
32	-64	0.37175982	0.23230132	0.89879405	-0.91667987	0.23245576	0.32505744
33	-66	0.34111805	0.22152465	0.91354546	-0.92626095	0.23908574	0.29133942
34	-68	0.31056294	0.20947735	0.92718385	-0.93494588	0.24432917	0.25725369
35	-70	0.28016650	0.19617469	0.93969262	-0.94273097	0.24816537	0.22287275
36	-72	0.25000000	0.18163563	0.95105652	-0.94961327	0.25057747	0.18826992
37	-74	0.22013378	0.16588270	0.96126170	-0.95559050	0.25155249	0.15351918
38	-76	0.19063706	0.14894199	0.97029573	-0.96066107	0.25108130	0.11869497
39	-78	0.16157773	0.13084307	0.97814760	-0.96482404	0.24915872	0.08387198
40	-80	0.13302222	0.11161890	0.98480775	-0.96807914	0.24578348	0.04912496
41	-82	0.10503527	0.09130577	0.99026807	-0.97042673	0.24095827	0.01452852
42	-84	0.07767979	0.06994319	0.99452190	-0.97186780	0.23468966	-0.01984305
43	-86	0.05101666	0.04757380	0.99756405	-0.97240394	0.22698819	-0.05391602
44	-88	0.02510460	0.02424323	0.99939083	-0.97203736	0.21786822	-0.08761734
45	-90	0.00000000	0.00000000	1.00000000	-0.97077086	0.20734801	-0.12087493
46	-92	-0.02424323	-0.02510460	0.99939083	-0.96860779	0.19544957	-0.15361778
47	-94	-0.04757380	-0.05101666	0.99756405	-0.96555210	0.18219866	-0.18577620
48	-96	-0.06994319	-0.07767979	0.99452190	-0.96160829	0.16762472	-0.21728197

phi deg	beta deg	u.e0	u.A0	u.c	u ICRS x	u ICRS y	u ICRS z
49	-98	-0.09130577	-0.10503527	0.99026807	-0.95678141	0.15176077	-0.24806855
50	-100	-0.11161890	-0.13302222	0.98480775	-0.95107706	0.13464332	-0.27807123
51	-102	-0.13084307	-0.16157773	0.97814760	-0.94450136	0.11631231	-0.30722732
52	-104	-0.14894199	-0.19063706	0.97029573	-0.93706098	0.09681096	-0.33547631
53	-106	-0.16588270	-0.22013378	0.96126170	-0.92876311	0.07618567	-0.36276002
54	-108	-0.18163563	-0.25000000	0.95105652	-0.91961544	0.05448594	-0.38902279
55	-110	-0.19617469	-0.28016650	0.93969262	-0.90962619	0.03176415	-0.41421159
56	-112	-0.20947735	-0.31056294	0.92718385	-0.89880409	0.00807550	-0.43827616
57	-114	-0.22152465	-0.34111805	0.91354546	-0.88715838	-0.01652215	-0.46116920
58	-116	-0.23230132	-0.37175982	0.89879405	-0.87469880	-0.04196848	-0.48284642
59	-118	-0.24179573	-0.40241567	0.88294759	-0.86143559	-0.06820080	-0.50326670
60	-120	-0.25000000	-0.43301270	0.86602540	-0.84737951	-0.09515427	-0.52239222
61	-122	-0.25690996	-0.46347783	0.84804810	-0.83254180	-0.12276206	-0.54018851
62	-124	-0.26252517	-0.49373803	0.82903757	-0.81693423	-0.15095549	-0.55662457
63	-126	-0.26684892	-0.52372049	0.80901699	-0.80056904	-0.17966427	-0.57167295
64	-128	-0.26988823	-0.55335287	0.78801075	-0.78345902	-0.20881662	-0.58530982
65	-130	-0.27165378	-0.58256342	0.76604444	-0.76561742	-0.23833953	-0.59751505
66	-132	-0.27215994	-0.61128123	0.74314483	-0.74705804	-0.26815888	-0.60827222
67	-134	-0.27142465	-0.63943640	0.71933980	-0.72779517	-0.29819969	-0.61756873
68	-136	-0.26946943	-0.66696025	0.69465837	-0.70784361	-0.32838629	-0.62539577
69	-138	-0.26631929	-0.69378546	0.66913061	-0.68721869	-0.35864253	-0.63174837
70	-140	-0.26200263	-0.71984631	0.64278761	-0.66593628	-0.38889195	-0.63662542
71	-142	-0.25655121	-0.74507881	0.61566148	-0.64401273	-0.41905805	-0.64002965
72	-144	-0.25000000	-0.76942088	0.58778525	-0.62146495	-0.44906440	-0.64196766
73	-146	-0.24238713	-0.79281257	0.55919290	-0.59831038	-0.47883490	-0.64244986
74	-148	-0.23375373	-0.81519615	0.52991926	-0.57456699	-0.50829397	-0.64149046
75	-150	-0.22414387	-0.83651630	0.50000000	-0.55025329	-0.53736674	-0.63910743
76	-152	-0.21360436	-0.85672028	0.46947156	-0.52538832	-0.56597923	-0.63532246
77	-154	-0.20218467	-0.87575801	0.43837115	-0.49999169	-0.59405857	-0.63016087
78	-156	-0.18993678	-0.89358230	0.40673664	-0.47408352	-0.62153318	-0.62365160
79	-158	-0.17691502	-0.91014888	0.37460659	-0.44768452	-0.64833294	-0.61582706
80	-160	-0.16317591	-0.92541658	0.34202014	-0.42081592	-0.67438939	-0.60672310
81	-162	-0.14877802	-0.93934743	0.30901699	-0.39349951	-0.69963590	-0.59637886
82	-164	-0.13378177	-0.95190676	0.27563736	-0.36575762	-0.72400784	-0.58483673
83	-166	-0.11824930	-0.96306329	0.24192190	-0.33761316	-0.74744276	-0.57214217
84	-168	-0.10224427	-0.97278921	0.20791169	-0.30908956	-0.76988053	-0.55834364
85	-170	-0.08583165	-0.98106026	0.17364818	-0.28021080	-0.79126349	-0.54349241
86	-172	-0.06907761	-0.98785583	0.13917310	-0.25100141	-0.81153662	-0.52764249
87	-174	-0.05204925	-0.99315894	0.10452846	-0.22148646	-0.83064769	-0.51085043
88	-176	-0.03481448	-0.99695636	0.06975647	-0.19169154	-0.84854734	-0.49317518
89	-178	-0.01744177	-0.99923861	0.03489950	-0.16164278	-0.86518926	-0.47467795
90	-180	-0.00000000	-1.00000000	0.00000000	-0.13136683	-0.88053025	-0.45542203
91	-182	0.01744177	-0.99923861	-0.03489950	-0.10089083	-0.89453041	-0.43547261
92	-184	0.03481448	-0.99695636	-0.06975647	-0.07024245	-0.90715313	-0.41489660
93	-186	0.05204925	-0.99315894	-0.10452846	-0.03944982	-0.91836529	-0.39376250
94	-188	0.06907761	-0.98785583	-0.13917310	-0.00854156	-0.92813726	-0.37214012
95	-190	0.08583165	-0.98106026	-0.17364818	0.02245325	-0.93644300	-0.35010051
96	-192	0.10224427	-0.97278921	-0.20791169	0.05350509	-0.94326013	-0.32771564
97	-194	0.11824930	-0.96306329	-0.24192190	0.08458402	-0.94856996	-0.30505831
98	-196	0.13378177	-0.95190676	-0.27563736	0.11565968	-0.95235756	-0.28220190
99	-198	0.14877802	-0.93934743	-0.30901699	0.14670132	-0.95461177	-0.25922017
100	-200	0.16317591	-0.92541658	-0.34202014	0.17767784	-0.95532520	-0.23618710
101	-202	0.17691502	-0.91014888	-0.37460659	0.20855778	-0.95449430	-0.21317664

phi deg	beta deg	u.e0	u.A0	u.c	u ICRS x	u ICRS y	u ICRS z
102	-204	0.18993678	-0.89358230	-0.40673664	0.23930938	-0.95211932	-0.19026253
103	-206	0.20218467	-0.87575801	-0.43837115	0.26990058	-0.94820428	-0.16751811
104	-208	0.21360436	-0.85672028	-0.46947156	0.30029907	-0.94275702	-0.14501612
105	-210	0.22414387	-0.83651630	-0.50000000	0.33047230	-0.93578909	-0.12282848
106	-212	0.23375373	-0.81519615	-0.52991926	0.36038753	-0.92731578	-0.10102612
107	-214	0.24238713	-0.79281257	-0.55919290	0.39001184	-0.91735602	-0.07967877
108	-216	0.25000000	-0.76942088	-0.58778525	0.41931219	-0.90593234	-0.05885478
109	-218	0.25655121	-0.74507881	-0.61566148	0.44825545	-0.89307081	-0.03862095
110	-220	0.26200263	-0.71984631	-0.64278761	0.47680842	-0.87880095	-0.01904232
111	-222	0.26631929	-0.69378546	-0.66913061	0.50493790	-0.86315565	-0.00018200
112	-224	0.26946943	-0.66696025	-0.69465837	0.53261070	-0.84617106	0.01789899
113	-226	0.27142465	-0.63943640	-0.71933980	0.55979370	-0.82788650	0.03514188
114	-228	0.27215994	-0.61128123	-0.74314483	0.58645389	-0.80834435	0.05149035
115	-230	0.27165378	-0.58256342	-0.76604444	0.61255841	-0.78758990	0.06689062
116	-232	0.26988823	-0.55335287	-0.78801075	0.63807459	-0.76567126	0.08129164
117	-234	0.26684892	-0.52372049	-0.80901699	0.66297004	-0.74263921	0.09464524
118	-236	0.26252517	-0.49373803	-0.82903757	0.68721263	-0.71854705	0.10690622
119	-238	0.25690996	-0.46347783	-0.84804810	0.71077057	-0.69345045	0.11803248
120	-240	0.25000000	-0.43301270	-0.86602540	0.73361250	-0.66740730	0.12798517
121	-242	0.24179573	-0.40241567	-0.88294759	0.75570745	-0.64047755	0.13672878
122	-244	0.23230132	-0.37175982	-0.89879405	0.77702498	-0.61272306	0.14423119
123	-246	0.22152465	-0.34111805	-0.91354546	0.79753519	-0.58420738	0.15046385
124	-248	0.20947735	-0.31056294	-0.92718385	0.81720875	-0.55499563	0.15540175
125	-250	0.19617469	-0.28016650	-0.93969262	0.83601702	-0.52515430	0.15902359
126	-252	0.18163563	-0.25000000	-0.95105652	0.85393202	-0.49475106	0.16131178
127	-254	0.16588270	-0.22013378	-0.96126170	0.87092655	-0.46385458	0.16225248
128	-256	0.14894199	-0.19063706	-0.97029573	0.88697421	-0.43253435	0.16183568
129	-258	0.13084307	-0.16157773	-0.97814760	0.90204945	-0.40086047	0.16005520
130	-260	0.11161890	-0.13302222	-0.98480775	0.91612764	-0.36890351	0.15690873
131	-262	0.09130577	-0.10503527	-0.99026807	0.92918511	-0.33673424	0.15239779
132	-264	0.06994319	-0.07767979	-0.99452190	0.94119919	-0.30442353	0.14652780
133	-266	0.04757380	-0.05101666	-0.99756405	0.95214830	-0.27204208	0.13930798
134	-268	0.02424323	-0.02510460	-0.99939083	0.96201196	-0.23966028	0.13075141
135	-270	0.00000000	-0.00000000	-1.00000000	0.97077086	-0.20734801	0.12087493
136	-272	-0.02510460	0.02424323	-0.99939083	0.97840688	-0.17517443	0.10969914
137	-274	-0.05101666	0.04757380	-0.99756405	0.98490318	-0.14320784	0.09724833
138	-276	-0.07767979	0.06994319	-0.99452190	0.99024423	-0.11151547	0.08355040
139	-278	-0.10503527	0.09130577	-0.99026807	0.99441583	-0.08016328	0.06863680
140	-280	-0.13302222	0.11161890	-0.98480775	0.99740518	-0.04921585	0.05254245
141	-282	-0.16157773	0.13084307	-0.97814760	0.99920091	-0.01873616	0.03530565
142	-284	-0.19063706	0.14894199	-0.97029573	0.99979314	0.01121456	0.01696795
143	-286	-0.22013378	0.16588270	-0.96126170	0.99917347	0.04057699	-0.00242591
144	-288	-0.25000000	0.18163563	-0.95105652	0.99733507	0.06929386	-0.02282818
145	-290	-0.28016650	0.19617469	-0.93969262	0.99427267	0.09731014	-0.04418819
146	-292	-0.31056294	0.20947735	-0.92718385	0.98998263	0.12457312	-0.06645249
147	-294	-0.34111805	0.22152465	-0.91354546	0.98446293	0.15103258	-0.08956500
148	-296	-0.37175982	0.23230132	-0.89879405	0.97771325	0.17664092	-0.11346716
149	-298	-0.40241567	0.24179573	-0.88294759	0.96973491	0.20135326	-0.13809807
150	-300	-0.43301270	0.25000000	-0.86602540	0.96053098	0.22512757	-0.16339469
151	-302	-0.46347783	0.25690996	-0.84804810	0.95010624	0.24792475	-0.18929196
152	-304	-0.49373803	0.26252517	-0.82903757	0.93846724	0.26970875	-0.21572302
153	-306	-0.52372049	0.26684892	-0.80901699	0.92562227	0.29044665	-0.24261936
154	-308	-0.55335287	0.26988823	-0.78801075	0.91158139	0.31010870	-0.26991101

phi deg	beta deg	u.e0	u.A0	u.c	u ICRS x	u ICRS y	u ICRS z
155	-310	-0.58256342	0.27165378	-0.76604444	0.89635645	0.32866845	-0.29752675
156	-312	-0.61128123	0.27215994	-0.74314483	0.87996106	0.34610276	-0.32539425
157	-314	-0.63943640	0.27142465	-0.71933980	0.86241063	0.36239186	-0.35344030
158	-316	-0.66696025	0.26946943	-0.69465837	0.84372234	0.37751941	-0.38159102
159	-318	-0.69378546	0.26631929	-0.66913061	0.82391515	0.39147250	-0.40977203
160	-320	-0.71984631	0.26200263	-0.64278761	0.80300978	0.40424167	-0.43790863
161	-322	-0.74507881	0.25655121	-0.61566148	0.78102870	0.41582096	-0.46592606
162	-324	-0.76942088	0.25000000	-0.58778525	0.75799614	0.42620786	-0.49374965
163	-326	-0.79281257	0.24238713	-0.55919290	0.73393801	0.43540334	-0.52130503
164	-328	-0.81519615	0.23375373	-0.52991926	0.70888195	0.44341179	-0.54851834
165	-330	-0.83651630	0.22414387	-0.50000000	0.68285725	0.45024103	-0.57531643
166	-332	-0.85672028	0.21360436	-0.46947156	0.65589486	0.45590225	-0.60162702
167	-334	-0.87575801	0.20218467	-0.43837115	0.62802732	0.46040997	-0.62737895
168	-336	-0.89358230	0.18993678	-0.40673664	0.59928878	0.46378195	-0.65250231
169	-338	-0.91014888	0.17691502	-0.37460659	0.56971488	0.46603920	-0.67692867
170	-340	-0.92541658	0.16317591	-0.34202014	0.53934278	0.46720581	-0.70059125
171	-342	-0.93934743	0.14877802	-0.30901699	0.50821110	0.46730893	-0.72342507
172	-344	-0.95190676	0.13378177	-0.27563736	0.47635984	0.46637867	-0.74536718
173	-346	-0.96306329	0.11824930	-0.24192190	0.44383036	0.46444796	-0.76635678
174	-348	-0.97278921	0.10224427	-0.20791169	0.41066531	0.46155248	-0.78633537
175	-350	-0.98106026	0.08583165	-0.17364818	0.37690858	0.45773053	-0.80524697
176	-352	-0.98785583	0.06907761	-0.13917310	0.34260524	0.45302291	-0.82303821
177	-354	-0.99315894	0.05204925	-0.10452846	0.30780149	0.44747278	-0.83965848
178	-356	-0.99695636	0.03481448	-0.06975647	0.27254454	0.44112553	-0.85506008
179	-358	-0.99923861	0.01744177	-0.03489950	0.23688263	0.43402862	-0.86919835
180	-360	-1.00000000	0.00000000	-0.00000000	0.20086488	0.42623148	-0.88203176
181	-362	-0.99923861	-0.01744177	0.03489950	0.16454126	0.41778528	-0.89352204
182	-364	-0.99695636	-0.03481448	0.06975647	0.12796250	0.40874285	-0.90363426
183	-366	-0.99315894	-0.05204925	0.10452846	0.09118002	0.39915843	-0.91233697
184	-368	-0.98785583	-0.06907761	0.13917310	0.05424585	0.38908759	-0.91960222
185	-370	-0.98106026	-0.08583165	0.17364818	0.01721253	0.37858701	-0.92540564
186	-372	-0.97278921	-0.10224427	0.20791169	-0.01986692	0.36771429	-0.92972658
187	-374	-0.96306329	-0.11824930	0.24192190	-0.05693917	0.35652783	-0.93254804
188	-376	-0.95190676	-0.13378177	0.27563736	-0.09395056	0.34508659	-0.93385681
189	-378	-0.93934743	-0.14877802	0.30901699	-0.13084728	0.33344996	-0.93364346
190	-380	-0.92541658	-0.16317591	0.34202014	-0.16757540	0.32167755	-0.93190238
191	-382	-0.91014888	-0.17691502	0.37460659	-0.20408098	0.30982901	-0.92863176
192	-384	-0.89358230	-0.18993678	0.40673664	-0.24031017	0.29796386	-0.92383362
193	-386	-0.87575801	-0.20218467	0.43837115	-0.27620926	0.28614130	-0.91751381
194	-388	-0.85672028	-0.21360436	0.46947156	-0.31172482	0.27442005	-0.90968196
195	-390	-0.83651630	-0.22414387	0.50000000	-0.34680375	0.26285813	-0.90035147
196	-392	-0.81519615	-0.23375373	0.52991926	-0.38139338	0.25151273	-0.88953945
197	-394	-0.79281257	-0.24238713	0.55919290	-0.41544160	0.24044001	-0.87726671
198	-396	-0.76942088	-0.25000000	0.58778525	-0.44889686	0.22969494	-0.86355766
199	-398	-0.74507881	-0.25655121	0.61566148	-0.48170837	0.21933112	-0.84844028
200	-400	-0.71984631	-0.26200263	0.64278761	-0.51382609	0.20940065	-0.83194598
201	-402	-0.69378546	-0.26631929	0.66913061	-0.54520088	0.19995391	-0.81410960
202	-404	-0.66696025	-0.26946943	0.69465837	-0.57578455	0.19103950	-0.79496922
203	-406	-0.63943640	-0.27142465	0.71933980	-0.60552999	0.18270398	-0.77456613
204	-408	-0.61128123	-0.27215994	0.74314483	-0.63439119	0.17499184	-0.75294466
205	-410	-0.58256342	-0.27165378	0.76604444	-0.66232338	0.16794528	-0.73015212
206	-412	-0.55335287	-0.26988823	0.78801075	-0.68928307	0.16160412	-0.70623859
207	-414	-0.52372049	-0.26684892	0.80901699	-0.71522816	0.15600568	-0.68125686

phi deg	beta deg	u.e0	u.A0	u.c	u ICRS x	u ICRS y	u ICRS z
208	-416	-0.49373803	-0.26252517	0.82903757	-0.74011798	0.15118463	-0.65526222
209	-418	-0.46347783	-0.25690996	0.84804810	-0.76391340	0.14717294	-0.62831238
210	-420	-0.43301270	-0.25000000	0.86602540	-0.78657688	0.14399972	-0.60046722
211	-422	-0.40241567	-0.24179573	0.88294759	-0.80807255	0.14169120	-0.57178873
212	-424	-0.37175982	-0.23230132	0.89879405	-0.82836626	0.14027056	-0.54234077
213	-426	-0.34111805	-0.22152465	0.91354546	-0.84742566	0.13975792	-0.51218891
214	-428	-0.31056294	-0.20947735	0.92718385	-0.86522025	0.14017029	-0.48140026
215	-430	-0.28016650	-0.19617469	0.93969262	-0.88172145	0.14152142	-0.45004331
216	-432	-0.25000000	-0.18163563	0.95105652	-0.89690262	0.14382188	-0.41818770
217	-434	-0.22013378	-0.16588270	0.96126170	-0.91073918	0.14707891	-0.38590406
218	-436	-0.19063706	-0.14894199	0.97029573	-0.92320856	0.15129647	-0.35326383
219	-438	-0.16157773	-0.13084307	0.97814760	-0.93429033	0.15647519	-0.32033903
220	-440	-0.13302222	-0.11161890	0.98480775	-0.94396619	0.16261237	-0.28720210
221	-442	-0.10503527	-0.09130577	0.99026807	-0.95222003	0.16970196	-0.25392569
222	-444	-0.07767979	-0.06994319	0.99452190	-0.95903795	0.17773461	-0.22058247
223	-446	-0.05101666	-0.04757380	0.99756405	-0.96440827	0.18669765	-0.18724495
224	-448	-0.02510460	-0.02424323	0.99939083	-0.96832161	0.19657517	-0.15398525
225	-450	-0.00000000	-0.00000000	1.00000000	-0.97077086	0.20734801	-0.12087493
226	-452	0.02424323	0.02510460	0.99939083	-0.97175119	0.21899383	-0.08798481
227	-454	0.04757380	0.05101666	0.99756405	-0.97126012	0.23148718	-0.05538477
228	-456	0.06994319	0.07767979	0.99452190	-0.96929746	0.24479955	-0.02314356
229	-458	0.09130577	0.10503527	0.99026807	-0.96586535	0.25889945	0.00867138
230	-460	0.11161890	0.13302222	0.98480775	-0.96096827	0.27375253	0.03999409
231	-462	0.13084307	0.16157773	0.97814760	-0.95461301	0.28932160	0.07076028
232	-464	0.14894199	0.19063706	0.97029573	-0.94680864	0.30556682	0.10090745
233	-466	0.16588270	0.22013378	0.96126170	-0.93756657	0.32244572	0.13037515
234	-468	0.18163563	0.25000000	0.95105652	-0.92690046	0.33991341	0.15910501
235	-470	0.19617469	0.28016650	0.93969262	-0.91482623	0.35792264	0.18704103
236	-472	0.20947735	0.31056294	0.92718385	-0.90136204	0.37642395	0.21412960
237	-474	0.22152465	0.34111805	0.91354546	-0.88652823	0.39536582	0.24031971
238	-476	0.23230132	0.37175982	0.89879405	-0.87034733	0.41469479	0.26556308
239	-478	0.24179573	0.40241567	0.88294759	-0.85284399	0.43435565	0.28981425
240	-480	0.25000000	0.43301270	0.86602540	-0.83404494	0.45429156	0.31303070
241	-482	0.25690996	0.46347783	0.84804810	-0.81397895	0.47444423	0.33517300
242	-484	0.26252517	0.49373803	0.82903757	-0.79267680	0.49475407	0.35620485
243	-486	0.26684892	0.52372049	0.80901699	-0.77017120	0.51516039	0.37609320
244	-488	0.26988823	0.55335287	0.78801075	-0.74649673	0.53560154	0.39480833
245	-490	0.27165378	0.58256342	0.76604444	-0.72168982	0.55601511	0.41232391
246	-492	0.27215994	0.61128123	0.74314483	-0.69578864	0.57633808	0.42861707
247	-494	0.27142465	0.63943640	0.71933980	-0.66883306	0.59650704	0.44366844
248	-496	0.26946943	0.66696025	0.69465837	-0.64086459	0.61645835	0.45746221
249	-498	0.26631929	0.69378546	0.66913061	-0.61192629	0.63612833	0.46998614
250	-500	0.26200263	0.71984631	0.64278761	-0.58206268	0.65545342	0.48123160
251	-502	0.25655121	0.74507881	0.61566148	-0.55131971	0.67437041	0.49119358
252	-504	0.25000000	0.76942088	0.58778525	-0.51974463	0.69281660	0.49987066
253	-506	0.24238713	0.79281257	0.55919290	-0.48738596	0.71072997	0.50726506
254	-508	0.23375373	0.81519615	0.52991926	-0.45429336	0.72804938	0.51338255
255	-510	0.22414387	0.83651630	0.50000000	-0.42051757	0.74471475	0.51823250
256	-512	0.21360436	0.85672028	0.46947156	-0.38611030	0.76066722	0.52182777
257	-514	0.20218467	0.87575801	0.43837115	-0.35112418	0.77584934	0.52418471
258	-516	0.18993678	0.89358230	0.40673664	-0.31561264	0.79020525	0.52532308
259	-518	0.17691502	0.91014888	0.37460659	-0.27962981	0.80368080	0.52526597
260	-520	0.16317591	0.92541658	0.34202014	-0.24323046	0.81622378	0.52403977

phi deg	beta deg	u.e0	u.A0	u.c	u ICRS x	u ICRS y	u ICRS z
261	-522	0.14877802	0.93934743	0.30901699	-0.20646988	0.82778402	0.52167405
262	-524	0.13378177	0.95190676	0.27563736	-0.16940380	0.83831356	0.51820144
263	-526	0.11824930	0.96306329	0.24192190	-0.13208829	0.84776681	0.51365759
264	-528	0.10224427	0.97278921	0.20791169	-0.09457966	0.85610068	0.50808102
265	-530	0.08583165	0.98106026	0.17364818	-0.05693438	0.86327469	0.50151299
266	-532	0.06907761	0.98785583	0.13917310	-0.01920897	0.86925116	0.49399741
267	-534	0.05204925	0.99315894	0.10452846	0.01854009	0.87399523	0.48558069
268	-536	0.03481448	0.99695636	0.06975647	0.05625644	0.87747508	0.47631156
269	-538	0.01744177	0.99923861	0.03489950	0.09388395	0.87966194	0.46624101
270	-540	0.00000000	1.00000000	0.00000000	0.13136683	0.88053025	0.45542203
271	-542	-0.01744177	0.99923861	-0.03489950	0.16864966	0.88005772	0.44390956
272	-544	-0.03481448	0.99695636	-0.06975647	0.20567755	0.87822540	0.43176022
273	-546	-0.05204925	0.99315894	-0.10452846	0.24239619	0.87501775	0.41903224
274	-548	-0.06907761	0.98785583	-0.13917310	0.27875194	0.87042273	0.40578520
275	-550	-0.08583165	0.98106026	-0.17364818	0.31469193	0.86443179	0.39207993
276	-552	-0.10224427	0.97278921	-0.20791169	0.35016413	0.85703998	0.37797826
277	-554	-0.11824930	0.96306329	-0.24192190	0.38511743	0.84824591	0.36354289
278	-556	-0.13378177	0.95190676	-0.27563736	0.41950174	0.83805185	0.34883719
279	-558	-0.14877802	0.93934743	-0.30901699	0.45326806	0.82646365	0.33392499
280	-560	-0.16317591	0.92541658	-0.34202014	0.48636854	0.81349081	0.31887042
281	-562	-0.17691502	0.91014888	-0.37460659	0.51875655	0.79914644	0.30373773
282	-564	-0.18993678	0.89358230	-0.40673664	0.55038678	0.78344725	0.28859106
283	-566	-0.20218467	0.87575801	-0.43837115	0.58121529	0.76641351	0.27349428
284	-568	-0.21360436	0.85672028	-0.46947156	0.61119955	0.74806903	0.25851081
285	-570	-0.22414387	0.83651630	-0.50000000	0.64029855	0.72844108	0.24370341
286	-572	-0.23375373	0.81519615	-0.52991926	0.66847283	0.70756037	0.22913402
287	-574	-0.24238713	0.79281257	-0.55919290	0.69568451	0.68546095	0.21486357
288	-576	-0.25000000	0.76942088	-0.58778525	0.72189740	0.66218013	0.20095178
289	-578	-0.25655121	0.74507881	-0.61566148	0.74707699	0.63775845	0.18745703
290	-580	-0.26200263	0.71984631	-0.64278761	0.77119053	0.61223949	0.17443614
291	-582	-0.26631929	0.69378546	-0.66913061	0.79420708	0.58566986	0.16194423
292	-584	-0.26946943	0.66696025	-0.69465837	0.81609750	0.55809900	0.15003458
293	-586	-0.27142465	0.63943640	-0.71933980	0.83683453	0.52957915	0.13875841
294	-588	-0.27215994	0.61128123	-0.74314483	0.85639279	0.50016515	0.12816481
295	-590	-0.27165378	0.58256342	-0.76604444	0.87474883	0.46991432	0.11830052
296	-592	-0.26988823	0.55335287	-0.78801075	0.89188115	0.43888634	0.10920984
297	-594	-0.26684892	0.52372049	-0.80901699	0.90777020	0.40714309	0.10093450
298	-596	-0.26252517	0.49373803	-0.82903757	0.92239840	0.37474847	0.09351350
299	-598	-0.25690996	0.46347783	-0.84804810	0.93575018	0.34176828	0.08698303
300	-600	-0.25000000	0.43301270	-0.86602540	0.94781195	0.30827001	0.08137634
301	-602	-0.24179573	0.40241567	-0.88294759	0.95857213	0.27432270	0.07672368
302	-604	-0.23230132	0.37175982	-0.89879405	0.96802115	0.23999674	0.07305214
303	-606	-0.22152465	0.34111805	-0.91354546	0.97615143	0.20536371	0.07038564
304	-608	-0.20947735	0.31056294	-0.92718385	0.98295738	0.17049618	0.06874481
305	-610	-0.19617469	0.28016650	-0.93969262	0.98843540	0.13546752	0.06814696
306	-612	-0.18163563	0.25000000	-0.95105652	0.99258388	0.10035171	0.06860600
307	-614	-0.16588270	0.22013378	-0.96126170	0.99540313	0.06522318	0.07013240
308	-616	-0.14894199	0.19063706	-0.97029573	0.99689541	0.03015657	0.07273318
309	-618	-0.13084307	0.16157773	-0.97814760	0.99706492	-0.00477344	0.07641184
310	-620	-0.11161890	0.13302222	-0.98480775	0.99591769	-0.03949235	0.08116841
311	-622	-0.09130577	0.10503527	-0.99026807	0.99346165	-0.07392598	0.08699937
312	-624	-0.06994319	0.07767979	-0.99452190	0.98970655	-0.10800074	0.09389773
313	-626	-0.04757380	0.05101666	-0.99756405	0.98466391	-0.14164376	0.10185299

phi deg	beta deg	u.e0	u.A0	u.c	u ICRS x	u ICRS y	u ICRS z
314	-628	-0.02424323	0.02510460	-0.99939083	0.97834701	-0.17478311	0.11085119
315	-630	-0.00000000	0.00000000	-1.00000000	0.97077086	-0.20734801	0.12087493
316	-632	0.02510460	-0.02424323	-0.99939083	0.96195210	-0.23926896	0.13190345
317	-634	0.05101666	-0.04757380	-0.99756405	0.95190903	-0.27047800	0.14391264
318	-636	0.07767979	-0.06994319	-0.99452190	0.94066151	-0.30090880	0.15687513
319	-638	0.10503527	-0.09130577	-0.99026807	0.92823093	-0.33049694	0.17076037
320	-640	0.13302222	-0.11161890	-0.98480775	0.91464015	-0.35918000	0.18553469
321	-642	0.16157773	-0.13084307	-0.97814760	0.89991345	-0.38689775	0.20116140
322	-644	0.19063706	-0.14894199	-0.97029573	0.88407649	-0.41359233	0.21760090
323	-646	0.22013378	-0.16588270	-0.96126170	0.86715621	-0.43920838	0.23481079
324	-648	0.25000000	-0.18163563	-0.95105652	0.84918083	-0.46369321	0.25274596
325	-650	0.28016650	-0.19617469	-0.93969262	0.83017975	-0.48699693	0.27135875
326	-652	0.31056294	-0.20947735	-0.92718385	0.81018350	-0.50907257	0.29059906
327	-654	0.34111805	-0.22152465	-0.91354546	0.78922368	-0.52987624	0.31041449
328	-656	0.37175982	-0.23230132	-0.89879405	0.76733289	-0.54936723	0.33075049
329	-658	0.40241567	-0.24179573	-0.88294759	0.74454467	-0.56750811	0.35155053
330	-660	0.43301270	-0.25000000	-0.86602540	0.72089347	-0.58426485	0.37275621
331	-662	0.46347783	-0.25690996	-0.84804810	0.69641451	-0.59960692	0.39430747
332	-664	0.49373803	-0.26252517	-0.82903757	0.67114378	-0.61350733	0.41614273
333	-666	0.52372049	-0.26684892	-0.80901699	0.64511797	-0.62594277	0.43819910
334	-668	0.55335287	-0.26988823	-0.78801075	0.61837435	-0.63689362	0.46041250
335	-670	0.58256342	-0.27165378	-0.76604444	0.59095079	-0.64634403	0.48271788
336	-672	0.61128123	-0.27215994	-0.74314483	0.56288562	-0.65428196	0.50504940
337	-674	0.63943640	-0.27142465	-0.71933980	0.53421760	-0.66069921	0.52734060
338	-676	0.66696025	-0.26946943	-0.69465837	0.50498586	-0.66559147	0.54952459
339	-678	0.69378546	-0.26631929	-0.66913061	0.47522983	-0.66895829	0.57153426
340	-680	0.71984631	-0.26200263	-0.64278761	0.44498918	-0.67080313	0.59330244
341	-682	0.74507881	-0.25655121	-0.61566148	0.41430373	-0.67113332	0.61476213
342	-684	0.76942088	-0.25000000	-0.58778525	0.38321345	-0.66996007	0.63584665
343	-686	0.79281257	-0.24238713	-0.55919290	0.35175834	-0.66729841	0.65648983
344	-688	0.81519615	-0.23375373	-0.52991926	0.31997841	-0.66316720	0.67662625
345	-690	0.83651630	-0.22414387	-0.50000000	0.28791361	-0.65758904	0.69619136
346	-692	0.85672028	-0.21360436	-0.46947156	0.25560377	-0.65059024	0.71512170
347	-694	0.87575801	-0.20218467	-0.43837115	0.22308854	-0.64220074	0.73335511
348	-696	0.89358230	-0.18993678	-0.40673664	0.19040738	-0.63245402	0.75083083
349	-698	0.91014888	-0.17691502	-0.37460659	0.15759945	-0.62138706	0.76748976
350	-700	0.92541658	-0.16317591	-0.34202014	0.12470359	-0.60904020	0.78327457
351	-702	0.93934743	-0.14877802	-0.30901699	0.09175828	-0.59545705	0.79812989
352	-704	0.95190676	-0.13378177	-0.27563736	0.05880158	-0.58068438	0.81200248
353	-706	0.96306329	-0.11824930	-0.24192190	0.02587109	-0.56477200	0.82484136
354	-708	0.97278921	-0.10224427	-0.20791169	-0.00699609	-0.54777263	0.83659799
355	-710	0.98106026	-0.08583165	-0.17364818	-0.03976340	-0.52974174	0.84722639
356	-712	0.98785583	-0.06907761	-0.13917310	-0.07239486	-0.51073744	0.85668329
357	-714	0.99315894	-0.05204925	-0.10452846	-0.10485511	-0.49082032	0.86492822
358	-716	0.99695636	-0.03481448	-0.06975647	-0.13710944	-0.47005326	0.87192370
359	-718	0.99923861	-0.01744177	-0.03489950	-0.16912380	-0.44850131	0.87763530
360	-720	1.00000000	-0.00000000	-0.00000000	-0.20086488	-0.42623148	0.88203176

Appendix D. Canonical formula and source ledger

ID	Formula	Definition
O01	$A0 = \hat{c} \times e0$	construct positive inner axis and right-handed device triad
O02	$R0 = [e0 \ A0 \ \hat{c}]$	device-to-ICRS orientation at zero phase
O03	$\phi = \omega_B(t - t0)$	positive outer phase
O04	$R_B = R(\hat{c}, \phi)$	active outer rotation
O05	$e = R_B e0$	transported bar-plane reference
O06	$A = R_B A0$	transported positive inner axis
O07	$\beta = -2\phi$	signed inner phase
O08	$\omega_{B_vec} = +\omega_B \hat{c}$	outer angular velocity
O09	$\omega_{A_vec} = -2\omega_B A(\phi)$	inner angular velocity
O10	$u = R(\hat{c}, \phi) R(A0, -2\phi) e0$	ordered active compound rotation
O11	$u = \cos(2\phi)e + \sin(2\phi)\hat{c}$	equivalent direct path
O12	$r_{plus/minus} = r_C \pm R u$	permanently labeled endpoints
O13	$u_\phi(0) = A0 + 2\hat{c}$	zero-phase sign gate
O14	$RPM = \omega_B 60 / (2\pi)$	rate conversion

Variable ledger

Symbol	Meaning	Units/type
$t0$	zero-phase epoch	time datum
r_{C0}	Earth-center device origin at datum	m, ICRS
$e0$	positive bar direction and +mass direction at $t0$	unit vector
$A0$	positive inner-axis direction at $t0$	unit vector
$\hat{c}$	positive outer-axis direction and Earth-CMB direction	unit vector
ϕ	positive outer phase	radian
β	signed inner phase $= -2\phi$	radian
R_B	active outer rotation matrix	SO(3)
u	positive endpoint direction	unit vector
R	endpoint radius	m
ω_B	positive outer angular speed	rad/s
ω_{A_signed}	signed inner angular speed $= -2\omega_B$	rad/s

Source ledger

#	Source	Use
1	K-Field_dual_axis_kinetic_mass_analysis_1784741897(1).json	authoritative ICRS apparatus vectors, rates, endpoint radius, Cell3 geometry, vertices, face planes, and 27-node registry
2	K-Field_dual_axis_kinetic_mass_CMB_traversal_lateral_force_computation_1784803951.json	continuity with the canonical dual-axis traversal and lateral-force code stack
3	K-Field_dual_axis_kinetic_mass_starting_orientation_1784805531.json	complete machine-readable starting-orientation contract and one-degree orientation ledger

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